

SPOT

Lucrative Problems

PROBE

The Opportunity

Ramesh Raskar



Preface

Everyone is trapped in the same paradox: one must innovate to survive, but innovation involves risk, and risk-reduction is a key tenet of institutions. The question is: **How to harness the freedom of creative thought to transcend what has come before while maintaining the discipline that prevents wasteful journeys down primrose paths?**

I've been there myself. I have faced this dilemma during my years of experience at the MIT Media Lab, as a Professor and Director of research group, as well as launching innovation teams at Facebook, contributing to special projects at Google X and Apple, and mentoring hundred+ startups at my MIT Venture class and via the AI venture studio C10Labs. I have come up with a set of practices that rationalize, potentiate, and discipline innovation at all phases, mapping out a general pathway by which you can identify, evaluate and pitch a new initiative. These “new rules” both encourage and discipline the creative process in order to maximize the chances of success—and thus of course minimize the risk of failure—and to properly communicate with key decision-makers.

I call it the “Spot-Probe” process. This process rationalizes all phases of innovation but especially idea generation (Spot) and idea validation (Probe).

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Introduction: The Innovator's Mindset

The Problem

Why is innovation so difficult? After all, every human being has the innate capacity to invent solutions to problems. It's our defining species characteristic: we call it "intelligence."

Intelligence may be understood as making progress faster than natural selection ever could. Instead of a slow, blind process of trial and error via the sorting of randomly generated options, we instead use our brains to leapfrog to a planned, foreseen solution. That's how a species that hasn't had the time to biologically evolve during the past ten thousand years has moved from the rudiments of agriculture to space travel and all the rest.

Moreover, in the current phase of civilization, one marked by the digital revolution, we have an ever-increasing number of tools, techniques, and technologies that both directly aid the innovation process and serve as frontiers for exploration. Almost off the top of one's head one can make an impressive list: AI, AR/VR, blockchain, IoT, 3D printing, synthetic biology, quantum computing, robotics...

As in a burgeoning biological ecosystem, every new added element increases the total number of interactions within the system and thus creates ever more options for further adaptation. Niches are created faster than they can be filled. In biology, this is called "adaptive radiation." It is most obvious after some cataclysm, whether a local one like a forest fire or

a global one like an asteroid impact: lots of new niches are created and filled.

So, we should be in a golden age of innovation, and in some obvious sense we are. But we are also told that 95 percent of product innovations never succeed and that only six percent of global executives are satisfied with the performance of their innovation initiatives. That sounds like the kind of efficiency we'd expect from natural selection! How can this be?

Surely one reason is that true innovation—the generation of a new and valuable idea which is then “made flesh”—is innately difficult. And so it is: any problem worthy of an innovative solution is almost by definition an entrenched problem reflecting the inertia of traditional thinking and institutional structures.

But that's always been the case. The newer, additional reason is that the success of past innovations burdens the present. The very proliferation of options for how and what to innovate and the ever-increasing complexity of the socioeconomic ecosystem can itself be overwhelming. In adaptive radiations, as evidenced in the fossil record, evolution occurs more quickly. The same holds true in the socioeconomic version: change accelerates in all dimensions.

The problem is that an ever-expanding plethora of options daunts. In such a seeming chaos, the rules governing what to do and how to do it seem to be up in the air. Without such rules, people often fall back on a sort of faith that an idea will surely find its niche somewhere in that ever-widening, constantly shifting socioeconomic ecosystem of the near future. They simply hope for the best and surrender to a selection-like sorting, holding tight to their idea and presuming all (especially decision-makers) will see its superiority—will be *converted*, that is, instead of *convinced*.

Decision-makers are a bit wary of the charismatic mode and far more insistent on being rationally convinced.

Thus, even if one can transcend the noise of the current moment and come up with a true innovation, convincing institutions to back it presents another challenge. Would-be founders cannot expect decision-makers to fall as deeply and instantly in love with their idea as they have. Not all ideas that die deserve to; often they are starved unnecessarily because of a lack of clear communication. After all, everyone is trapped in the same paradox: one must innovate to survive, but innovation involves risk, and risk-reduction is a key tenet of institutions. **The question is: How to harness the freedom of creative thought to transcend what has come before while maintaining the discipline that prevents wasteful journeys down primrose paths?**

The Solution

I've been there myself. After years of experience at the MIT Media Lab, as Director of the Camera Culture research group, as well as launching innovation teams at Facebook, contributing to special projects at Google X and Apple, and mentoring hundred + startups at my MIT Venture class and via the venture studio C10Labs, I have come up with a set of practices that rationalize, potentiate, and discipline innovation at all phases, mapping out a general pathway by which you can identify, evaluate and pitch a new initiative. These “new rules” both encourage and discipline the creative process in order to maximize the chances of success—and thus of course minimize the risk of failure—and to properly communicate with key decision-makers.

I call it the “Spot-Probe” process. This process rationalizes all phases of innovation but especially idea generation (Spot) and idea validation (Probe). This document will explicate the

first step, Spot, but for context, the process as a whole is as follows:

1. **Spot:** identify a “grand opportunity” to solve an important problem
2. **Probe:** justify the solution
3. **Grow:** build out the solution
4. **Launch:** release the solution into the wild

Spot itself is divided into three phases, each of which contains steps that will be explained in some detail in the body of this document:

*Phase 1: **Identify** and specify a “grand opportunity” to solve an important problem*

*Phase 2: **Analyze** both the opportunity and solution*

*Phase 3: **Communicate** the solution to decision-makers*

But rules, steps, and practices are not enough in and of themselves. They are the “letter of the law,” as it were. In order to succeed, you also need to imbibe the “spirit of the law.” you need to adopt an innovator’s mindset and habits of thought. No rule set in and of itself can guarantee success without a higher dedication to the point of it all. So it is with that mindset we begin.

The Innovator’s Mindset

The innovator’s mindset is most fundamentally the mastering of the essential tension between the freedom to create, to think outside the box, and the discipline required to fully interrogate one’s beloved creation in order to prove its worth. **You need to have both the arrogance to believe you can**

see beyond others and the humility to ruthlessly seek out weaknesses in your own ideas. Both require courage. They are also different modes of thinking purposely deployed at different points of the process. Spot-Probe codifies these shifts in thinking; it is a template for innovation. As you will see even in the Spot phase of the process, with which this document is concerned, there is purposeful oscillation between free and disciplined thinking. We will return to aspects of the innovator's mindset as we review the steps in the Probe process.

System 1 versus System 2 Thinking

One way to conceptualize these styles of thought is System 1 versus System 2 thinking, popularized in Daniel Kahneman's work. System 1 thinking is enthusiastic, "romantic," charismatic, unbounded. This is what you want to generate a new idea: a clear-sky, "what if?" attitude. System 2 thinking is calm, realistic, critical, relentless. This is what you want to interrogate a new idea: a clear-minded, "can it work?" attitude. If you misapply either style of thinking at the wrong time or in the wrong context you risk failure: you neither want to kill your new idea in the crib (or prevent its birth) by overly critical thinking nor do you want to protect your idea from the reality it will have to withstand to survive. Even in the Spot phase of the process, I have built in System 2 thinking—just enough to convince yourself and others to move on to the more System 2-based Probe phase which follows. It's so easy to become either toxically positive or toxically hypercritical: perhaps the main value of Spot-Probe is that strictly following the method removes both temptations. It is a template for innovation.

Habits of Thought for Innovators: Interrogating the Problem

It's not just balancing System 1 and System 2 thinking. Multifaceted thinking even dominates the search for an idea.

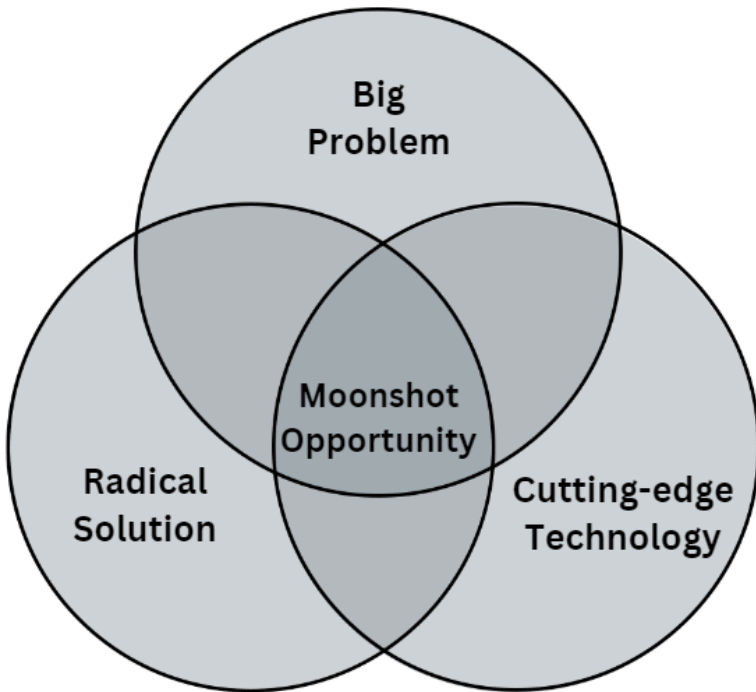
The “habit of thought” you should inculcate is interrogating the problem you’re interested in from all sides. Part of the challenge is becoming familiar enough with all aspects of the problem to be able to locate gaps—the niches you will aim to fill with a new product. We will say more about this old-fashioned research below.

But the key beyond familiarity is finding a real unsolved opportunity—a “grand opportunity”—in your problem area. Remember, a **grand challenge is not always a grand opportunity**. It’s a bit of a Goldilocks problem. You want to frame your problem and its solution properly: not too big, not too small. Innovators tend to get overly passionate about either the general problem they want to solve (e.g. climate or mental health) or the specific solution they first intuit (e.g. a consumer app or an online marketplace). To avoid both pitfalls, you want to carefully and explicitly determine the scope of both your problem and solution. You’re looking for the biggest possible impact to the largest problem you can manage to solve in the simplest way. The most efficient lever, as it were.

Habits of Thought for Innovators: Exploring Intersections

Remember, technological innovations do not occur in a vacuum. They survive if they can integrate themselves into the pre-existing socio-technological system, even if they alter that system. Biological metaphors can help here: no new species, however impressively constructed, will survive unless it can integrate itself into what already exists, into the available ecosystem. Moreover, in an ever-splintering world of contemporary technoscience, you will almost certainly have to avail yourself of many fields of expertise—and experts, too. Whether building a team or researching an opportunity, you will have to range widely. (Note how this is an instance of how cross-fertilization is the mother of innovation.)

Your goal is the proverbial “moonshot”—something that lies at the intersection of a big problem, cutting-edge technology, and the vision to see a radical solution.



Interestingly, one of the founders of modern technoscience, the English lawyer and philosopher Francis Bacon, suggested right at the outset of the scientific revolution that there should be a class of people who synthesize knowledge and find applications. They do not do the original research themselves; they are applied scientists and technologists who bring progress out of the lab and into the world. This is pretty close to your role as a technological innovator.

In fact, not being a specialist sometimes allows a person to cast their vision at a higher level, across domains of knowledge. The twentieth-century historian and philosopher Isaiah Berlin once separated thinkers into two categories, “hedgehogs” and “foxes,” taken from an ancient Greek epigram, “A fox knows many things but a hedgehog knows one big thing.” Hedgehogs specialize and burrow their way into their field whereas foxes range across fields, hunting for connections. Foxes are generalists; hedgehogs are specialists. Each has their place, but you’ll want to lean in the direction of the fox as an innovator, even if your team consists of a few hedgehogs! You want to be a generalist, as much as possible, and you’ll want your team, both as a group and as much as they can as individuals, to range as widely as possible across all dimensions of your problem and solution. A more System 1 kind of exercise.

That said, you need to get up to speed (if you’re not already) on whatever emerging technologies apply via podcasts, white papers, scholarly journal articles, conferences, and your social network. Expose yourself to all kinds of seemingly far-flung projects and domains—remember, Darwin and Wallace found inspiration from economics, of all things! Even your team should be built in such a way as to maximize the chances of cross-fertilization. You’ll want audacious, intelligent, and determined people, of course, but you’ll also want as wide a variety of backgrounds, knowledge, and personalities as you can manage.

With the proper mental attitude in mind, let’s turn now to the rules of the game: how Spot works, step by step. First stop: discovering a grand opportunity to solve an entrenched problem.

Phase 1:

Recognizing a

Grand Opportunity

Spot begins with recognizing a grand opportunity in any domain. In this phase, your goal is to identify a pressing unsolved problem and then to generate as many potential approaches as possible.

Your tasks:

1. Define the **domain** of the grand opportunity
2. Create a **SMART Question**

Defining a Domain and a Question

First, you must define a domain, such as transportation, medical care, or communication. Within that domain, you will drill down to an emerging technology opportunity. It might be helpful to think of the process in cinematic terms: first, you set up a wide, establishing shot to define the setting; next, you move in closer to isolate where the action will take place.

Take transportation, for example. Self-driving cars are an emerging technology that solve certain problems but at the same time create more, much as a new organism fills a niche as it creates others. Self-driving cars could create even more traffic, could endanger safety, eliminate many jobs, and disrupt some companies' revenues. Cities will need to find new ways to regulate driving—how, for example, does a driver's license apply to self-driving cars?

So what's the opportunity? In other words, how might the ecosystem evolve? What additional emerging technologies can solve these subproblems? Most importantly, what lucrative innovations and business opportunities will arise? Those are what might be called “moonshots” or “grand opportunities”—problems since time immemorial the solution to which has yet to be found and the effects of which will literally change daily life for billions.

Grand Opportunity Domains

Examples of domains where potential solutions effect billions of people.



- **Transportation and Vehicles**
Technologies: Sensors, AI/ML, Robots, Big Data, Neural Networks
- **Health and Wellbeing**
Technologies: Big Data, Medical Imaging, BioBanks, AR ect...
- **Infrastructure & Environment**
Technologies: Sensors, AI/ML, Robots, Big Data, Neural Networks

Domain defining of course takes place at least in part well before you sit down and formally start the Spot process. All of

your learning and research, including, crucially, competitive research, flows into your definition of a domain, a problem, and a potential solution. In fact, many of you will have already come up with an idea and will be using the Spot-Probe process as a kind of proof of concept. This is fine, but if you're in the latter category, make sure you are truly interrogating your problem and plan! Keep the spirit of the law in mind as you go through the steps, that is.

Write down your grand opportunity topic and then we will convert that into a SMART question.

Have a look at some good and bad examples of grand opportunity topics below.



Grand Opportunity Topics

Good Examples

- ✓ How can we use AI and imaging to improve the safety and cost of autonomous vehicles?
- ✓ How can we share sensor data for better localization and tracking of the area around you?

Bad Examples

- ✗ How can we improve the experience of self-driving cars?
- ✗ What if self-driving cars are low cost and safe?

Good topics...

- are specific!
- solve a significant problem and generate a profound impact.
- are feasible and actionable.

Creating a SMART Question

Let's stick with the autonomous driving ecosystem. We've got our domain and we have our problem in tight focus. Now we want to generate a specific question that will naturally lead to an array of subproblems and solutions, and see what are the best, most exciting and lucrative intersections of those

problems and solutions. (Alternatively, you could be using Spot-Probe to validate or interrogate an idea you already have. But you will have to convert the idea into a SMART question.)

In order to discipline the process, we'll take advantage of the well-known SMART Question method. SMART Questions are (ideally) one sentence that is simultaneously:

Specific

Measurable

Achievable

Relevant

Time-bound

Note that SMART Questions share and encourage the mindset that Spot-Probe aims to inculcate: a mental template for the production of innovation that balances creativity and discipline. The different styles of thought from the last section (Systems 1 and 2) are embedded into the process. They check and balance each other. As one sits around on one's own or with a team, ideas are generated. This is free thinking. Then, a small set of ideas (by vote or otherwise) are given the SMART treatment to see if they can pass this first System-2-thinking test. At this idea-generation stage, a light dose of System 2 thinking is all you need: just enough to lower the risk of self-delusion, of "falling in love with an idea," but not enough to needlessly kill the idea in its crib. In this sense, Spot is not unlike agile methods of physical production, with rapid prototyping and lower costs of both initiating—and if necessary abandoning—dead ends.

Note the Goldilocks nature of the question and a very broad general solution:

"How can we build a business that exploits emerging self-driving cars?"

"How can we use AI and imaging to improve safety for self-driving cars?"

This is a good start. The first focuses too much on business. The second one combines a problem (safety) and solution (AI) for an emerging sector of self-driving cars, but both the problem and solution are too specific. Let's broaden that a bit so that the Question becomes a SMART question and is more open to exploration.

"How can we create an ecosystem in the next five years to achieve a safe, livable, green city that builds on emerging self-driving cars?"

You didn't say, "technology"—you chose a specific "hammer," self-driving cars. You didn't say "make transportation better," you chose an actionable, measurable goal, "improve transportation safety and create sustainable cities." You created (or, rather, recognized) a "nail," that is: a problem your hammer can solve. The problems are longstanding and huge: traffic, safety, livability. It's a global problem and it's been around for well over a century. No one's actually fully figured it out. The scope of impact is gigantic. This is a moonshot, which is different from pie-in-the-sky. There is in fact a moon up there in the sky; there is no pie.



SMART Question

“How can we create an ecosystem in the next five years to achieve a safe, livable, green city that builds on emerging self-driving cars?”

Toxic Positivity and a Corrective for Charisma

Once you've identified your SMART question for the grand opportunity, it's time to bring in a little more discipline. While you will Probe far more deeply, at this point, to keep yourself grounded, it's time to analyze your big new idea. **Remember that the key pitfall is a kind of toxic positivity around an idea—and the self-defeating domination of charismatic, overly extroverted leadership or founder-worship.**

In fact, as we move through the second phase of Spot, pay attention to how the nature of the exercises ensure that extroverted charismatic types are partially undermined through group dynamics. Spot-Probe is a corrective for charisma, and another blow, incidentally, struck for introverts. Nothing wrong with extroverts, of course, but, decision-makers rightly want to be convinced via argument not seduced via charisma.

How SMART is the question? "How can we create an ecosystem in the next five years to achieve a safe, livable, green city that builds on emerging self-driving cars? You can try to rank whether the question you have created is SMART by assigning a 1 to 5 score (5 is the best) to each of the five

criteria. A combined score of 15 out 25 is good enough at this stage.

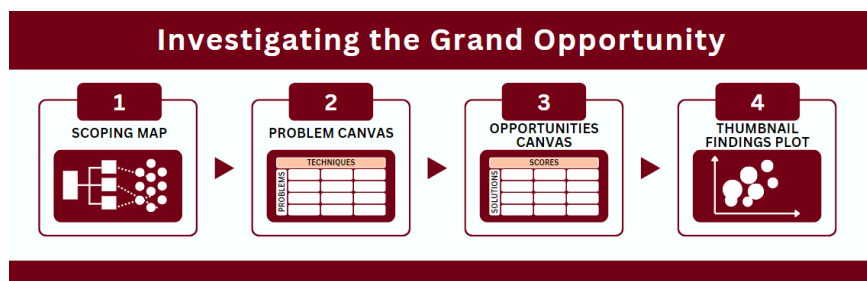
- Specific: It is sufficiently specific about the domain, but the target problem is somewhat open ended and the solution is not too prescriptive. (4 out of 5)
- Measurable: We mentioned what to measure: safe, livable and green but did not specify how to measure. Should it be 20% better or make it safer by 30%? We left it open. (2 out of 5)
- Achievable: We did pick a question for which solution does not already exist. Are there technical, regulatory or societal factors that would likely kill the idea? We can imagine the solution appearing in the near term. (4 out of 5)
- Relevant: Problems related to smart cities and urban population are very lucrative. It is not too niche. But maybe we have chosen a city-wide ecosystem that is too complex and ill-defined. (3 out 5)
- Time-bound: We are saying 5 years. It could be because we guess that is roughly how long it could take for self-driving cars to be seen on the streets. Or it could be because a startup or a corporate venture needs to fund and show results in a finite time. (4 out of 5)

We can do better by making it measurable and choosing an ecosystem that is less complex than a city-network. But that is fine. If we make the 'measurable' aspect too precise, we will lose the freedom to explore. A combined score of 17 out of 25 is sufficient to move on to the next phase of deeper analysis.

Phase 2: Analyzing the Grand Opportunity

The first step of Spot included a preliminary, built-in justification of your idea—what I call “the grand opportunity”—the SMART Question. It was intentionally just enough to keep you honest (that is, not starry-eyed) but not enough to undermine free creativity. Now we come to a bit more justification, a kind of dry run for the Probe phase that is outside the scope of this document. While Probe will analyze the grand opportunity in depth, the analysis phase of Spot provides a quick-and-dirty proof of concept to justify further investment of time and effort. Note that it’s a bit more

involved than a mere SMART Question, but not as involved as Probe will be.



Here's the roadmap:

S: A *Scoping Map* that captures the span of your project and the resources involved — the stakeholder, people and organizations at hand or that you'll need, the users and beneficiaries of the solution, the competitive landscape and the breakthroughs and risks of the project.

P: A *Problem Canvas* that lists all the options at the intersection of subproblems and opportunities inherent in the project, all the skills, knowledge, and techniques relevant to the SMART question, and all the associations among the former two lists.

O: A *Opportunities Canvas* that ranks in order of importance the options identified in the problem canvas associating techniques with expected impact and roughly enumerating cost, time, scope of impact, among other items.

T: A *Thumbnail Findings Plot* that visualizes the entire analysis and forms the basis for the one-pager you will use to

convince decision-makers of moving forward to the Probe phase.

Scoping Map

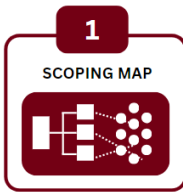
First, take the SMART Question and on a white board with post-its, set up a resource map. Use one post-it for each item as described below. Whatever your current institutional structure (from a tiny start-up to a unit within a large multinational) list all the relevant people and organizations you will need to marshal in order to design and develop your solution.

- Who are the organizations, companies, startups, government organizations that are related to the grand opportunity?
- Who are the main partners?
- Who are the major stakeholders, customers, interest groups, and venture capitalists?
- What kind of data and information are available about the opportunity?

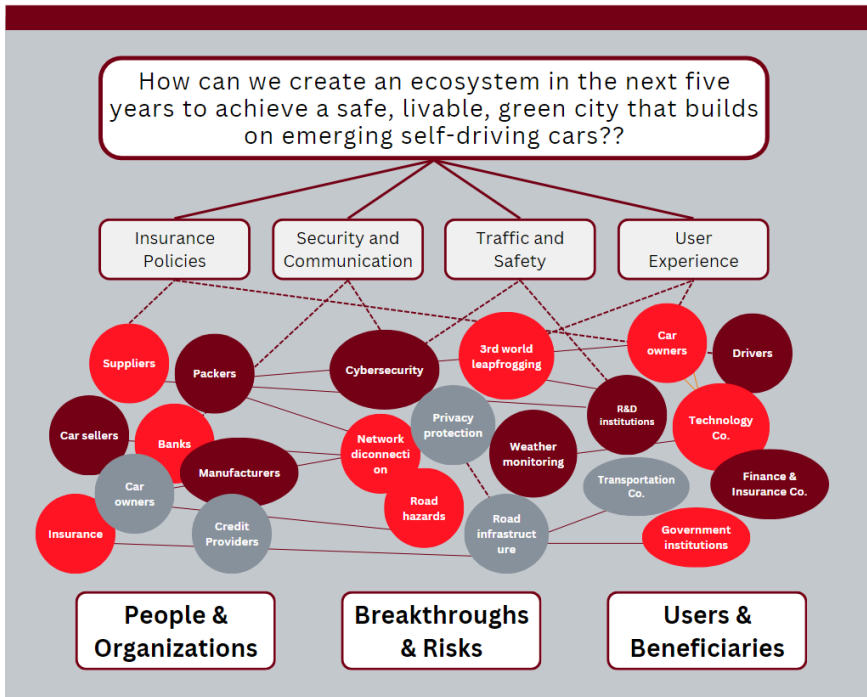
Then, list all the possible end-users and beneficiaries of your solution. Use cases obtain here, of course, as well as partners you may sign up and most importantly, who pays, ultimately.

- Who are the potential users who will benefit?
- Who are the main influencers?
- Who might pay? Who might be sponsors?

You may want to involve outside experts at this point, especially to identify where you may need further research. You'll want to graph users and use cases: you can use an interrelationship diagram (essentially a simple flowchart) to map the connections.



Plotting the Scoping Map



Finally, make a third list of all the trends and risks involved with your solution. What friction, if any, can you expect from regulatory agencies, both domestic and international? What are the financial or other risks? Are there limiting factors, such as time, distance, the lack of existing infrastructure, and the like? But don't forget the other side: what specific benefits will your new product or service provide? What will be its impact?

The most important output of the Scoping Map is the graph of stakeholders.

You're just making lists at this point—really, just items in post-it-able form so you can arrange and rearrange them with ease on your white board or table top. Don't worry about ranking in terms of importance. But you will want to draw (literally, on your white board) as many connections among your individual post-it items as you can. Doing so will lower the risk that you've left out something important from your lists.

"That sounds great," I can hear you thinking. "But how can I be sure I capture everything relevant?" One could brainstorm, yes, but that is a bit too free for this stage. We have an actual goal, however hazy at this stage. Thus, better to "brainswarm." **Brainswarming is essentially directed brainstorming—yet another dynamic interaction between free creativity and disciplined deduction.** It's not unlike how your smartphone finds pathways between where you currently are and where you want to go: you can work forward from where you are or backward from where you want to be. In this case, it's not a matter of geography but of project management. Encourage your team to think both forward and backward, along with the group nature of the exercise, this greatly reduces the risk of missing something important.

Another advantage of brainswarming is that work is done by individuals in silence and then brought back to the group. This undermines the dominance of extroverts, who, after all, tend to be the kind of people most likely to lead charismatically. Spot-Probe really is a dynamic equilibrium among various countervailing tendencies, in both people and ideas, and is above all rooted in the fruitful cross-fertilization of ideas that stand or fall on a rational basis. It's a check against charisma.

Problem Canvas

Use the list of items in your resource map to create a table that associates the problems and opportunities you found with the skills you can leverage to identify the “techniques” you will use to create your solution. Please note that my use of “technique” here means any tactic or strategy; any technical, operational process or mode of production; and even any mindset or social arrangement of production that will be necessary to institute in the design and development of your product or service.

The canvas will try to create a matrix of permutations and combinations of subproblems and solutions. This is a mixing process you have been driving haphazardly inside your head but you never had the **courage to explore the adjacent boxes** in this matrix.

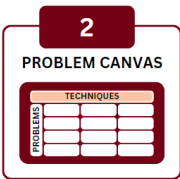
1. Make a list of all the subproblems associated with your opportunity.
2. Make a list of all the technologies or techniques that could be leveraged in creating solutions.
3. In a table, associate each problem with the technologies you listed in order to come up with options to approach creating solutions.
4. Upvote (or downvote) each option to narrow in on the top ten solutions.

The nature of this table template will help you identify gaps (ie, empty cells) the filling of which you can then focus on.

Once again, this is not a ranked list: the point is to capture everything you can. Once again, you can use brainstorming to generate as many possibilities as you can. One other advantage of brainstorming is that it maximizes the potential

of your hopefully diverse team by breaking down siloed thinking. It makes foxes out of hedgehogs, that is, at least at the collective level.

Here’s an example based on autonomous cars ecosystem:



Creating a Problem Canvas

Example: How can we create an ecosystem in the next five years to achieve a safe, livable, green city that builds on emerging self-driving cars?

		TECHNIQUES			
PROBLEMS		Sensors	Artificial Intelligence	Communications	Fincancing
	Auto Cybersecurity	Biometrics, Fingerprints	Abnormal behavior detection		
	Retrofitting Used Cars	lidar, radar, DIY package w/ CPU			New loans and leases
	Insurance Policies	Cameras and diagnostic data for risk prevention	actuary extension	Cloud services to collect data	Subsidized hardware / payment plans
	Safety for Pedestrians	camera, lidar, new beacon technology	pedestrian behavior		
	In-car Experience	Displays; audio, AR, VR - entertainment system	Specific content pushing, shopping		Special services, and insurance policies

The diagram categorizes problems like *Auto Cybersecurity* and *Retrofitting Used Cars* in the first column, with techniques that generate solutions such as *Sensors* and *Artificial Intelligence* across the top. Each cell illustrates a solution idea, for instance, the *DIY Sensor Package* employs *Sensors* to address the problem of *Retrofitting Used Cars*.

In your table of problems versus skills and techniques, the cells are the various options. By vote, possibly even silent

and anonymous, select the top ten cells to input into the next stage of the process, the Solution Canvas. (Note that voting, however done, is another check against those charismatic extroverts!)

If you already had an idea you were exploring, it will be visible as just one of the boxes in this problem-techniques matrix. In the next sections, your goal is to identify all the adjacent boxes, explore them and **objectively compare whether your original idea is better than the one in the adjacent boxes**. Will you be honest? You will have to systematically eliminate each option till the most lucrative one emerges.

Opportunity Canvas

It is difficult to be honest and dispassionate when one of your own ideas is being compared to others in those adjacent boxes. Now comes the ranking you have no doubt been waiting for! The opportunity canvas is a guide for determining the technology or technique that will create the solution with the most impact. Using your knowledge and best judgment, in another table associate the options you identified in the last step with the impacts you expect. Crucially, you will want to—somewhat roughly—enumerate the cost, time, and societal “reach” of each impact. Here, “reach” means the entire population, however subdivided, that would benefit from your product or service. The goal here is, again, moonshot-like: aim for a reach in the billions, literally.

Let's see the example of the autonomous cars ecosystem.



Creating the Opportunities Canvas

CONSTRAINTS & IMPACT

SOLUTIONS	Options	Cost	Time	Regulations	Sum of Constraints	Impact Quality	Impact Quantity	Impact Score (Qual * Qty)
	DIY Sensor Package	1	2	3	6	9	8	72
	Beacon Devices for People	9	8	5	22	7	5	35
	Displays for In-car Experiences	7	5	6	18	7	5	35
	Detection of Abnormal Behavior	4	4	4	12	5	9	45
	Risk Prediction for Insurance	1	1	6	8	3	6	18
	Processing Image Data for Analysis	3	3	6	12	3	9	27
	Subsidized hardware	10	7	7	24	8	8	64
	Cloud Services to Collect Data	2	8	6	16	6	9	54
	In-car Entertainment System	2	8	5	15	3	5	15
	Parking Availability	9	8	2	19	4	5	20

The potential solutions, such as DIY sensor package and Beacon Devices for people are listed in the first column. The subcategories of constraints and impact are shown across the top. Note that I use some very simple arithmetic operations here to calculate both “the sum of constraints” and “the impact score.” Don’t forget, we will soon Probe all of this; all we want to do now is make sure we haven’t gone all

starry-eyed and fooled ourselves into thinking we have a grand opportunity if in fact we don't.

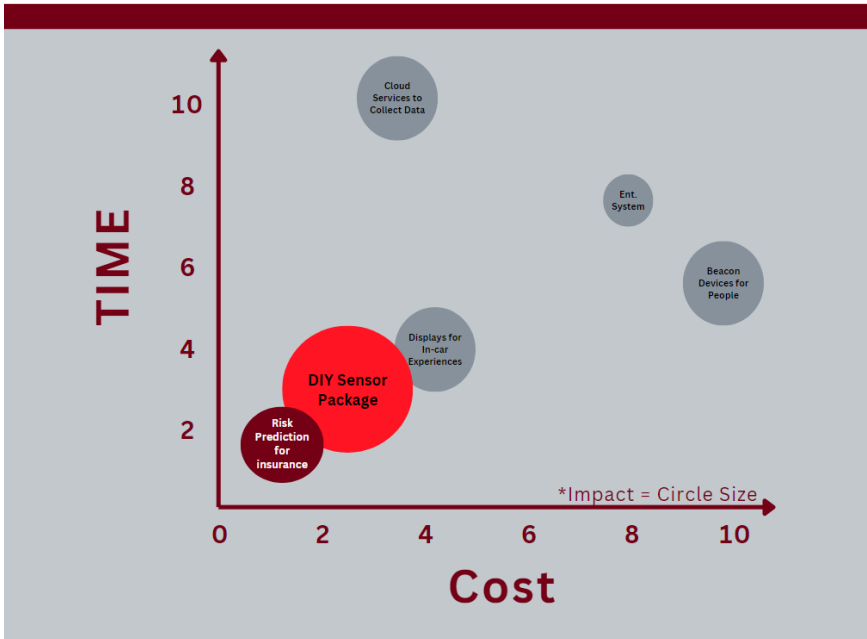
The enumeration here is qualitative; **avoid “analysis paralysis” in this stage.** Relative weights based on educated guesses are all that are needed. Simply rank from 1 to 10, sum the rows for impact quantity, assign an impact quality ranking, and multiply quantity by quality to get the impact score.

Thumbnail Findings Plot

Finally, create a thumbnail findings plot, which is an overlay of a word cloud on a simple graph that plots time (the y-axis) against cost (the x-axis). Impact is the size of the circle.



Creating the Thumbnail Findings Plot



This nifty little graph takes the various options you've isolated and roughly quantified in the solutions canvas and creates an instant visualization of the breadth of impact of each. The diameter of the circle represents the expected impact of each option, wherever it may lie in the time-vs-cost domain. Note how, for example, In constructing such a plot, you may well surprise yourself: your favored option may turn out to cover less literal area than you had thought. Here, we see that the DIY Sensor Package or a Beacon for pedestrians can be a lucrative opportunity in the autonomous car ecosystem. But one of them costs less versus the other.

This thumbnail plot forms the basis of your one-pager for decision-makers, the gatekeepers who will determine if your idea can move on. A picture is worth a thousand words, as they say. Business leaders, venture capitalists, foundation heads—the purse-holders, that is—have no time to waste but nevertheless want to be wowed by a new idea. It may have ever been thus but it does seem to be ever more pitched: important people simply have no time. Thus, this last step, much like the SMART Question, is neither a luxury nor a pointless act of martinet-like discipline. It's crucial.

Whenever I had to raise a lot of funds from investors or get a green light for my project from a VP at large companies like Facebook or Apple, it did not help to pitch a specific idea. Pitching almost never works. It is better to show a buffet of possibilities so the decision makers feel they are part of the decision making. The thumbnail shows that DIY Sensor Package and Beacon for Pedestrians have roughly the same impact but one of them costs less. This will allow the VP (or you) to make a quick decision that the Sensor Package may be the way to go.

What really works is creating a SMART Question that is exciting but is also broadly aligned with their goals. But more importantly the findings plot shows several potential options plotted on time-cost axis. It prompts decision-makers to begin considering their budgets and priorities.

Phase 3:

Summarizing the Grand Opportunity for Leaders

Business leaders need a one-page summary of the entire Spot process in order to judge not only the grand opportunity but also the efficiency of the team bringing that opportunity to their attention. You will want to define the following, all of which you can pull from the Spot documents and charts you will have already created:

- ✓ Why?—the grand opportunity and need
- ✓ Who?—the societal and business impact
- ✓ What?—the product or service's novelty
- ✓ How?—the plan to get it done

Summarizing the Findings



It's worth fleshing out this ready-made outline.

The *Why?* is simply the big idea itself:

- Why does this grand opportunity matter?
- How is the current situation addressed, if at all?
- What are the limitations of the current practices?

The *Who?* would be, less politely, “who cares?”—who cares if your idea comes to fruition? What is, in other words, the scope and nature of the expected impact? What difference will it make and to how many people? What populations will be positively affected?

The *What?* is probably best answered by your SMART Question. At least, SMART is a good place to start.

- What are you trying to achieve with your innovation?
- Why do you believe it will be successful?
- What, exactly, are you trying to accomplish?

You want to impress with your big idea, your novel solution. Be careful not to use any off-putting jargon: keep it simple, direct, concise, and forceful. If your idea is strong, you won't need much decoration, after all.

Finally, the *How?* question. This may well be where your audience leans in to really judge you. The following questions are based on the “Heilmeier Catechism” used at DARPA. Coming up with nifty ideas is great; proving that you can bring them to the marketplace is something else. You’ll want to map out users, resources, risks, workarounds, and milestones.

- What are the risks and payoffs?
- How much will it cost?
- How long will it take to implement the solution?
- What benchmarks will you set up in order to judge progress toward launch?
- What metrics will you use to measure success after launch?
- What are the potential risks and payoffs?

You will pull the answers from your mappings of potential problems, opportunity, and resources: all of Spot leads up to this presentation, after all.

You’ll probably also want to append your findings plot to your one-pager. Yes, you’ll want to hold yourself to a one-pager, but be prepared to expand in the happy event that your audience is interested. In fact, the various charts and graphs you will have produced in the analysis phase themselves provide evidence of your highly efficient thought process and management techniques! It wouldn’t hurt to have them on hand.

This summary serves as a comprehensive overview that can be presented to various stakeholders, including partners, customers, top management, colleagues, and investors. They can respond to the Thumbnail Findings Plot and the Summary.

When you ask for **feedback**, use the principle of 3/3/3: ask for three things they like, three they dislike, and three they would like to change.

* * *

And that's a high-level view of the Spot phase of the Spot-Probe process!

Note in conclusion that one overarching theme has been that cross-fertilization is often the basis of innovation. This is true beyond technological innovation. The classic example is the discovery of natural selection. Both Darwin and Wallace, each of whom independently came up with the idea, had read a work of economics by Malthus in which the interaction between limited resources and population growth were worked out in a sphere seemingly far removed from biology.

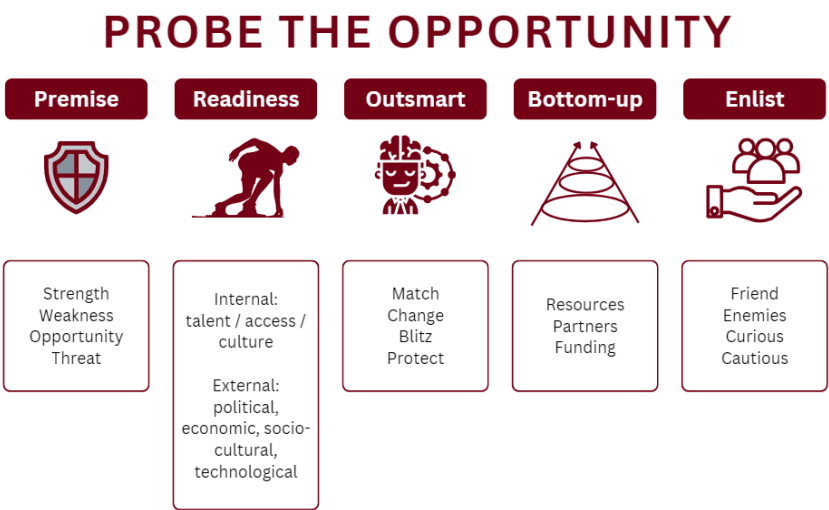
In fact, it's a good bet that originality may well be nothing more than cross-fertilization of the already-existing, all the more so today as our economic-technological ecosystem radiates in complexity with increasing speed. As you have seen, much of Spot consists of exploring possibly fruitful intersections whether among knowledge domains, markets, technologies and techniques, or individuals on your team. Innovation, that is, is an essentially social and collective process. Realizing this truth undermines the potential for charismatic leadership of a founding guru, a key pitfall in the current innovation process.

There is much more to say, of course, especially about Probe—all of which will be provided in my upcoming, full-length book. (But to tantalize you a bit, a Probe preview appendix concludes this booklet!)

To sum up, I recognized a widespread problem with innovation. I saw a grand opportunity to solve that problem, I created a landscape of current approaches and map of stakeholders, and through an iteration of problem and opportunity canvas landed on this process. I hope you will Spot-Probe every nascent idea in your head!

Appendix: Preview of PROBE

This booklet only covered SPOT part of the SPOT-PROBE process. Here is a teaser about the PROBE. During Probe you more rigorously validate your solution and adjust it to your own organization. (System 2 thinking dominates here!) Here are the phases of the Probe process followed by a quick explanation for each:



Premise: Put the SPOT output (ie, your premise) through a strength-weakness-opportunities-threats (SWOT) analysis.

Readiness: Evaluate your organization’s readiness with respect to both internal and external factors. The internal factors are talent, access, and culture (TAC); the external factors are political, economic, socio-cultural, and technological (PEST).

Outsmart: With your constellation of internal and external factors determined, and in the light of your SWOT analysis, choose one of

the following general strategies in order to “outsmart the future,” that is, to avoid pitfalls:

- *Match*: Pair up internal strengths with exploitable external opportunities.
- *Change*: Overcome weaknesses prior to or in parallel with the project—or somehow convert those weaknesses into advantages.
- *Blitz*: Build more strengths or marshal existing strengths in order to neutralize threats.
- *Protect*: Monitor emerging threats and prepare a Plan B.

Bottom-up blueprint: Build an internal roadmap that maps out resources, partners, and funding available internally. With these building blocks, you can create the blueprint. Instead of thinking hierarchically, think about a network dependency graph.

Enlist: Engage stakeholders to refine the roadmap in order get buy in

Glossary

System 1 Thinking: System 1 thinking is enthusiastic, “romantic,” charismatic, unbounded. This is what you want to generate a new idea: a clear-sky, “what if?” attitude.

System 2 Thinking: System 2 thinking is calm, realistic, critical, relentless. This is what you want to interrogate a new idea: a clear-minded, “can it work?” attitude.

grand opportunity: An unfilled niche in the ecosystem; your high-impact solution to an identified problem.

hedgehogs and foxes: The philosopher Isaiah Berlin’s handy distinction between specialists (hedgehogs) and generalists (foxes); based on a pre-Socratic fragment.

techniques: Any social, cultural, or technological practice that can be applied to a solution.

hammers and nails: A distinction between two types of techniques (as defined directly above)..

reach: Sociocultural or socioeconomic impact; the further the reach, the larger the impact.

analysis paralysis: The danger of taking analysis of a solution past the point of diminishing returns such that progress is unnecessarily slowed or stalled.

SMART Question: A riff off of the well-established SMART statement which forces consideration of actionable, doable solutions.

brainswarming: A form of group work that controls for the top-down, often extrovert-dominated features of brainstorming.

About the author

Ramesh Raskar is an Associate Professor at MIT Media Lab and directs the Camera Culture research group and the Program on Decentralized AI. He is co-founder of the AI Venture Studio C10Labs. His focus is on Machine Learning and Imaging for health and sustainability. They span research in physical (e.g., sensors, health-tech), digital (e.g., automated and privacy-aware machine learning) and global (e.g., geomaps, autonomous mobility) domains.

At MIT, his co-inventions include camera to see around corners, femto-photography, automated machine learning (auto-ML), private ML (split-learning), low-cost eye care devices (Netra, Catra, EyeSelfie), a novel CAT-Scan machine, motion capture (Prakash), long distance barcodes (Bokode), 3D interaction displays (BiDi screen), new theoretical models to augment light fields (ALF) to represent wave phenomena and algebraic rank constraints for 3D displays (HR3D).

He has worked on special research projects at Google [X], Facebook, Apple and co-founded/advised several startups. He launched REDX.io, a platform for young innovators to explore AI-for-Impact. He frequently consults for dynamic organizations to conduct 'SpotProbing' exercises to spot business venture problem and probe solutions.

He holds 100+ US patents.

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